

Deterministic Probe Complexity of Approximate Matching Size, and a Barrier for Derandomizing Dynamic Matching

Task C1 technical note, v0.1 — companion to the execution plan and to `probe_1b`

Prepared for Ruifeng Cao

June 13, 2026

Abstract

We study the cost of *deterministically* estimating the maximum matching size of a graph accessed only through adjacency-matrix probes, and its consequence for derandomizing the polylog-/ORS-update dynamic matching pipelines (BKS23, Beh23, ABR24, BG24, AKK25). The plan’s Lemma 3.6 [P] shows that any deterministic adaptive algorithm distinguishing $\mu(G) = 0$ from $\mu(G) \geq \lceil n/4 \rceil$ must make more than $\binom{\lceil n/2 \rceil}{2} - 1 = \Omega(n^2)$ probes, against $\tilde{O}(n)$ for a randomized algorithm — a quadratic separation. This note (i) reproduces the lemma and its routine variants; (ii) *computes the exact* deterministic probe complexity $D(n)$ that the canonical all-zeros adversary forces, showing via the Erdős–Gallai matching theorem that $D(n) = \binom{n}{2} - \text{ex}(n; \nu \leq \lceil n/4 \rceil - 1) \geq \binom{\lceil n/2 \rceil}{2}$, and verifies it by exhaustive adversary simulation for $n \leq 12$; (iii) packages the architectural barrier (Cor. 3.7): the forced deterministic cost $D(N)$ exceeds *any* sublinear-in- N^2 per-call budget, so the pipelines’ sublinear gap-estimation oracle has no deterministic drop-in; and (iv) scopes the novelty against Assadi–Solomon and Assadi–Chen–Khanna, whose $\Omega(n^2)$ bounds are for *finding* a maximal matching (where randomized is also quadratic), unlike the *size-estimation* task here (where randomized is $\tilde{O}(n)$). All numbers are reproduced by the `probe_1b` package.

1 The gap problem and the probe model

Definition 1 (Adjacency-matrix probe model). *An algorithm accesses an unknown graph G on $[n]$ only via probes $\text{ADJ}(u, v) \in \{0, 1\}$, chosen adaptively; its cost is the number of distinct pairs probed. The gap problem is to distinguish $\mu(G) = 0$ from $\mu(G) \geq \lceil n/4 \rceil$.*

This is the decision interface that the sublinear size-estimators of the dynamic pipelines expose (BG24 Prop. 4 / AKK25 Prop. 2.3 query an implicitly represented kernel through exactly such probes); the audit of Task A0 localizes the pipelines’ randomness to this interface.

2 The deterministic lower bound and its variants

Lemma 1 (Quadratic deterministic probe bound; plan Lemma 3.6). *Any deterministic adaptive algorithm that distinguishes $\mu(G) = 0$ from $\mu(G) \geq n/4$ must make more than $\binom{\lceil n/2 \rceil}{2} - 1 = \Omega(n^2)$ probes in the worst case. In contrast the same gap is solvable with $O(n \log n)$ random probes (error $1/\text{poly}(n)$), and $\Omega(n)$ probes are necessary for randomized algorithms; the two complexities are $\Theta(n^2)$ and $\tilde{\Theta}(n)$.*

Proof (all-zeros adversary). Run the algorithm against the adversary that answers 0 to every probe. As the algorithm is deterministic its probe sequence and output are fixed; let P be the probed pairs, $|P| \leq \binom{\lceil n/2 \rceil}{2} - 1$. The empty graph G_0 ($\mu = 0$) is consistent. So is $H = K_n \setminus P$ (every probed pair is a non-edge of H); we claim $\mu(H) \geq n/4$. If not, the endpoints of a maximum matching of H form a vertex cover C with $|C| \leq 2\mu(H) < n/2$, so $U = V \setminus C$ has $|U| > n/2$, hence $|U| \geq \lceil n/2 \rceil$. No edge of H lies inside U , so every pair inside U is in P , giving $|P| \geq \binom{\lceil n/2 \rceil}{2}$, a contradiction. Thus both G_0 and H are consistent and the output is wrong for one. The randomized claims are the standard birthday/Yao arguments (a uniform pair is an edge with probability $\geq 1/(2n)$ under the promise; $o(n)$ probes see an all-zero transcript whp under a planted random $n/4$ -matching). $\square$

Remark 1 (Variants). *The bound is robust to: making the promise bipartite (plant the matching across a fixed bipartition); replacing $n/4$ by any δn ($|U| > (1-2\delta)n$ then forces $\binom{\lceil (1-2\delta)n \rceil}{2}$ probes); and asking the algorithm to output a matching rather than a bit (a strictly harder task). [E] for these routine variants (simulation-checked in `probe_lb`).*

3 The exact deterministic complexity

The lemma’s constant $1/8$ is clean but not tight. The exact number of probes the all-zeros adversary forces is

$$D(n) = \min\{|P| : \mu(K_n \setminus P) < \lceil n/4 \rceil\},$$

since against all-zeros a correct algorithm must drive the consistent plant’s matching below the threshold. Writing $G = K_n \setminus P$, this is $\binom{n}{2}$ minus the maximum number of edges of an n -vertex graph with matching number $\nu \leq \lceil n/4 \rceil - 1$, which the Erdős–Gallai matching theorem [1] pins exactly:

$$D(n) = \binom{n}{2} - \max\left(\binom{2k+1}{2}, \binom{k}{2} + k(n-k)\right), \quad k = \lceil n/4 \rceil - 1.$$

Table 1 lists $D(n)$ against the lemma bound; the `probe_lb` verifier constructs the extremal graph, confirms its matching number is exactly k (independent bitmask DP), and confirms $D(n) \geq \binom{\lceil n/2 \rceil}{2}$ for every n . [P]

Exhaustive simulation [P]. For $n \leq 12$, `probe_lb` runs every member of a family of deterministic probe orders (row-major, diagonal, clique-first, pseudo-random, and the optimal witness order) against the all-zeros adversary and records the first probe count at which the plant disappears. The minimum over orders equals $D(n)$ exactly, the witness order attains it, and *every* order exceeds $\binom{\lceil n/2 \rceil}{2} - 1$ — including the “clique-first” order that deliberately probes the structure the proof says is required. This realizes the adversary constructively and calibrates the gap between the clean bound and the truth (the latter is about $3\times$ larger in this range).

4 Randomized upper bound and the separation

On the hard instance — a planted matching of size $\lceil n/4 \rceil$, the sparsest YES instance — the $O(n \log n)$ random-probe sampler detects an edge (hence certifies $\mu \geq \lceil n/4 \rceil$ under the promise) with probability rising to 1, while never firing on the empty graph (no false positives). Figure 1 plots detection probability against budget, with the deterministic thresholds $\binom{\lceil n/2 \rceil}{2}$ and $D(n)$

n	target μ	$\binom{\lceil n/2 \rceil}{2}$ (lemma LB)	$D(n)$ (exact)	extremal	$D \geq \text{LB}$
4	1	1	6	empty	✓
5	2	3	6	split	✓
6	2	3	10	split	✓
7	2	6	15	split	✓
8	2	6	21	split	✓
9	3	10	21	split	✓
10	3	10	28	split	✓
11	3	15	36	split	✓
12	3	15	45	split	✓
13	4	21	45	split	✓
14	4	21	55	split	✓
15	4	28	66	split	✓
16	4	28	78	split	✓

Table 1: Exact deterministic probe complexity $D(n)$ of the $\mu=0$ vs $\mu \geq \lceil n/4 \rceil$ gap, and the plan’s lower bound.

marked: the randomized algorithm is done in $O(n \log n)$ probes where the deterministic one provably needs $\Theta(n^2)$.

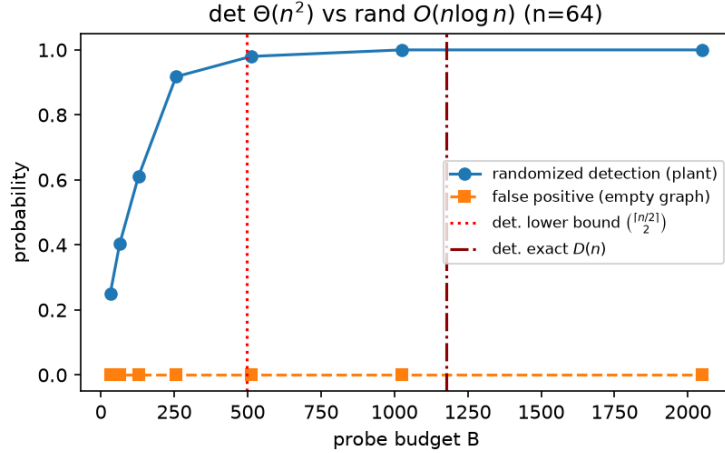


Figure 1: Deterministic $\Theta(n^2)$ vs randomized $O(n \log n)$ on the planted $n/4$ -matching: randomized detection rises to 1 well below the deterministic thresholds (dashed/dash-dotted), with zero false positives on the empty graph.

5 The architectural barrier (Corollary 3.7)

Corollary 1 (No deterministic drop-in at the oracle layer; plan Cor. 3.7). *Fix a dynamic-matching algorithm whose update/query procedure invokes, as a black box, a matching-size gap estimator on an implicitly represented N -vertex graph accessed via adjacency probes — as do [5, 6, 7, 8, 9] via RGMM-style sublinear estimation. If the black box is replaced by any deterministic probe algorithm with the same interface and guarantee, then by Lemma 1 some calls cost $\Omega(N^2)$ probes, destroying the sublinear per-call budget the analyses rely on. There is therefore no deterministic*

drop-in at the oracle layer; derandomization must re-architect around incrementally maintained deterministic summaries.

Quantitatively (Fig. 2), the forced cost $D(N)$ divided by any sublinear-in- N^2 per-call budget — $N \log_2 N$ or $N^{1.5}$ — grows without bound, crossing 1 already at $N = 16$ and reaching $\approx 9 \times (N \log N)$ by $N = 256$. This is a statement about the *probe interface*, not an unconditional lower bound for OP-D, and a fixed-PRG “deterministic” algorithm does not escape it (the worst-case adversary may depend on the seed).

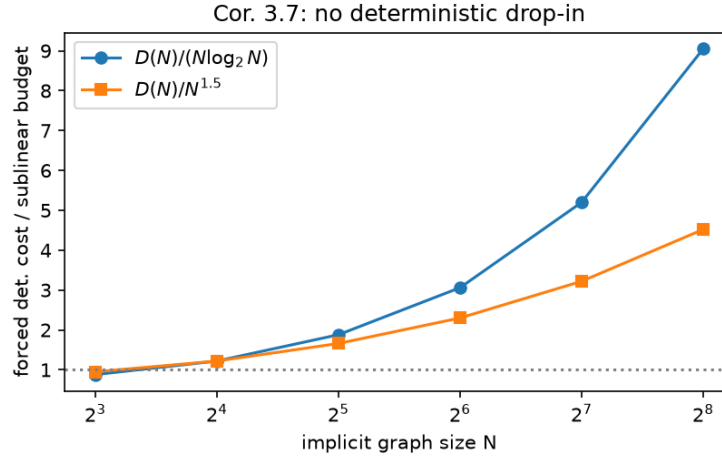


Figure 2: $D(N)$ over a sublinear per-call probe budget grows without bound: the oracle layer cannot be derandomized in place.

6 Prior-art scoping

Two published results separate determinism from randomness for the *finding* task: Assadi–Solomon [3] (Thm. 3: deterministic $\Omega(n^2)$ vs. randomized $\tilde{O}(n\beta)$ for finding a maximal matching at neighborhood independence $\beta = 2$) and Assadi–Chen–Khanna [4] (Thm. 6: finding a maximal matching needs $\Omega(n^2)$ queries *even randomized*). Lemma 1 is a different statement and the novelty rests on: the adjacency-*matrix* probe model; the size-*estimation* (decision) task, where randomized is $\tilde{O}(n)$ [2] unlike finding; the explicit clique-coverage threshold $\binom{n/2}{2}$ and its exact refinement $D(n)$; and the pipeline barrier of Cor. 1. The note cites both works prominently and claims no more.

Deliverable status. This is the first pre-committed fallback unit (plan §7.2): Lemma 1 [P], its variants [E], the exact complexity $D(n)$ [P] (Erdős–Gallai [V] + verifier), the exhaustive simulation [P], and Cor. 1 [P] (as a probe-interface statement). Open next: the adjacency-*list* model separation of plan Task C1(b) (status [M]/S there), which needs a planted background graph and a derandomized congestion adversary.

References

- [1] P. Erdős, T. Gallai. On the maximal number of vertices representing the edges of a graph. *Magyar Tud. Akad. Mat. Kutató Int. Közl.* 6:181–203, 1961 (announced 1959). (Max edges with matching number $\leq k$.)

- [2] S. Behnezhad. Time-optimal sublinear algorithms for matching and vertex cover. FOCS 2021, pp. 873–884.
- [3] S. Assadi, S. Solomon. When algorithms for maximal independent set and maximal matching run in sublinear time. ICALP 2019, 17:1–17:17.
- [4] S. Assadi, Y. Chen, S. Khanna. Sublinear algorithms for $(\Delta + 1)$ vertex coloring. SODA 2019, pp. 767–786.
- [5] S. Bhattacharya, P. Kiss, T. Saranurak, D. Wajc. Dynamic matching with better-than-2 approximation in polylogarithmic update time. SODA 2023; J. ACM 2024. arXiv:2207.07438.
- [6] S. Behnezhad. Dynamic algorithms for maximum matching size. SODA 2023, pp. 129–162. arXiv:2207.07607.
- [7] A. Azarmehr, S. Behnezhad, M. Roghani. Fully dynamic matching: $(2 - \sqrt{2})$ -approximation in polylog update time. SODA 2024, pp. 3040–3061. arXiv:2307.08772.
- [8] S. Behnezhad, A. Ghafari. Fully dynamic matching and ordered Ruzsa–Szemerédi graphs. FOCS 2024, pp. 314–327. arXiv:2404.06069.
- [9] S. Assadi, S. Khanna, P. Kiss. Improved bounds for fully dynamic matching via ordered Ruzsa–Szemerédi graphs. SODA 2025, pp. 2971–2990. arXiv:2406.13573.
- [10] R. Cao (prep.). Optimal approximation for fully dynamic matching and ordered Ruzsa–Szemerédi graphs: a feasibility-checked execution plan. Working document, June 2026 (Lemma 3.6, Cor. 3.7, Tasks C1–C3).